

REAL-TIME STOCK MARKET DATA VISUALIZATION

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for the partial fulfilment of the requirements to award the degree of

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In

**Computer Science and Engineering
School of Engineering and Sciences**

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Certificate

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This is to certify that the work present in this Project entitled **“REALTIME DATA ANALYSIS AND VISUALIZATION”** has been carried out by our group under my/our supervision. The work is genuine, original, and suitable for submission to the SRM University – AP for the award of Bachelor of Technology/Master of Technology in **School of Engineering and Sciences**.

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1. Introduction

This report documents the Real-Time Stock Data Analysis System, a comprehensive solution for monitoring and analyzing stock market data. The system fetches real-time stock data from the Alpha Vantage API, performs statistical analysis to detect anomalies, and presents the results through an interactive web dashboard.

Key capabilities:

- Real-time data collection from financial markets
- Statistical analysis and anomaly detection
- Interactive visualization of price and volume trends
- Fallback mechanisms for uninterrupted service

The system is designed for financial analysts, traders, and anyone interested in monitoring stock market movements with analytical insights.

2. System Architecture

The system follows a multi-threaded producer-consumer pattern with three main components:

1. **Data Producer:** Fetches stock data from Alpha Vantage API at regular intervals
2. **Data Analyzer:** Processes incoming data to detect statistical anomalies

3. **Web Server:** Serves the dashboard and provides data endpoints

Producer —► Analyzer —► Web Server

All components communicate through a shared queue and data structure with proper thread synchronization.

3. Key Components

3.1 Real Stock Producer

- Fetches data from Alpha Vantage API (GLOBAL_QUOTE endpoint)
- Handles API rate limits with fallback data generation
- Maintains realistic volume ranges for different stocks
- Implements error handling and retry logic

3.2 StockAnalyzer

- Maintains rolling window of historical data (default: 10 points)
- Calculates moving averages and standard deviations
- Implements z-score based anomaly detection
- Threshold configurable for sensitivity adjustment

3.3 Web Server

- Serves HTML dashboard with Plotly visualizations
- Provides JSON data endpoint at /data
- Handles concurrent requests

- Includes mock data for initial testing
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4. Data Flow

1. Producer fetches data every 15 seconds (configurable)
 2. Raw data placed in thread-safe queue
 3. Analyzer processes queued data:
 - Updates historical data windows
 - Calculates statistics
 - Detects anomalies
 4. Processed data stored in memory (last 100 points)
 5. Dashboard requests data via AJAX every 10 seconds
 6. Data formatted and sent as JSON
-

5. Technical Implementation

5.1 Threading Model

- Producer, analyzer, and server run in separate threads
- Queue provides thread-safe communication
- DataLock prevents race conditions on shared data

5.2 Statistical Methods

- Moving average: $\mu = \Sigma x/n$

- Standard deviation: $\sigma = \sqrt{(\sum(x-\mu)^2/n)}$
- Anomaly detection: $z\text{-score} = |x-\mu|/\sigma > \text{threshold (default=2.0)}$

5.3 Visualization

- Plotly.js for interactive charts
- Price chart with linear scale
- Volume chart with logarithmic scale
- Anomaly indicators in stats cards

6. Dashboard Features

1. Real-time Price Chart

- Shows price trends for all tracked stocks
- Time-based x-axis with auto-scaling

2. Volume Analysis

- Logarithmic scale accommodates large ranges
- Highlights unusual trading activity

3. Statistical Summary

- Current price and volume
- Mean and standard deviation
- Visual anomaly indicators

4. Connection Status

- Shows real-time connection health
- Automatic retry mechanism

7. Performance Considerations

- **Memory:** Limits stored data to last 100 points
- **CPU:** Efficient numpy calculations
- **Network:** 15-second API polling avoids rate limits
- **Fallback:** Mock data generation when API fails
- **Visual:** Plotly's react mode for efficient updates

8. Error Handling

- API failure detection and fallback
- Queue timeout prevents deadlock
- Thread monitoring in main loop
- Comprehensive logging
- User-friendly error messages

9. Security Considerations

1. **API Key:** Currently hardcoded (should be moved to environment variables)
2. **CORS:** Enabled for dashboard access

3. **Input Validation:** Not needed (no user input)
4. **HTTPS:** Recommended for production deployment

10. Future Enhancements

1. **Authentication:** Secure dashboard access
 2. **Alerting:** Email/SMS notifications for anomalies
 3. **Configuration:** External config file for symbols/intervals
 4. **Persistence:** Database storage for historical data
 5. **Advanced Analysis:** Machine learning models
 6. **Deployment:** Docker containerization
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Conclusion

The Real-Time Stock Data Analysis System provides a robust platform for monitoring stock market data with analytical insights. Its modular design allows for easy extension, and the interactive dashboard offers intuitive visualization of complex financial data. With proper production hardening, this system could serve as the foundation for professional financial analysis tools.